

ME23312 ENGINEERING THERMODYNAMICS

UNIT V – Gas Mixtures and Psychrometry • Lecture Notes

UNIT 5 • LECTURE NOTES

Unit V – Gas Mixtures and Psychrometry

Detailed lecture notes — composition and properties of ideal-gas mixtures, Dalton's and Amagat's laws, entropy of mixing, psychrometric properties and the psychrometric chart, adiabatic saturation, and the psychrometric processes of air conditioning, with solved numericals

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SYLLABUS

UNIT V — GAS MIXTURES AND PSYCHROMETRY (9 + 3 periods)

Mole and Mass fraction, Dalton's and Amagat's Law. Properties of gas mixture – Molar mass, gas constant, density, change in internal energy, enthalpy, entropy and Gibb's function. Psychrometric properties, Psychrometric charts. Property calculations of air vapour mixtures by using chart and expressions. Psychrometric process – adiabatic saturation, sensible heating and cooling, humidification, dehumidification, evaporative cooling and adiabatic mixing. Simple Applications.

Course Outcome CO5: Evaluate the properties of gas mixtures and the psychrometric properties for various psychrometric processes.

Text: P. K. Nag, *Engineering Thermodynamics*, 6th Edn., McGraw-Hill (Chapters 10 and 16).

References: Cengel & Boles, *Thermodynamics – An Engineering Approach*, 8th Edn.; R. K. Rajput, *A Text Book of Engineering Thermodynamics*, 5th Edn.; C. P. Arora, *Refrigeration and Air Conditioning*. Saturation properties of water from IAPWS-97.

CONTENTS

1. Introduction – mixtures of gases and moist air
2. Composition of gas mixtures – mass and mole fractions, Dalton's and Amagat's laws
3. Properties of ideal-gas mixtures – molar mass, gas constant, density, u , h , s and g (with worked examples)
4. Psychrometric properties of moist air and the psychrometric chart (with worked examples)
5. Psychrometric processes – sensible heating and cooling, humidification, dehumidification, evaporative cooling and adiabatic mixing (with worked examples)
6. Simple applications – summer and winter air conditioning, cooling towers and drying
7. Practice problems
8. Summary and formula sheet
9. Review questions (two-mark and thirteen-mark)

Learning objectives. After studying this unit you should be able to: (i) describe the composition of a gas mixture by mass and mole fractions and convert between them; (ii) state Dalton's and Amagat's laws and use them to find partial pressures and partial volumes; (iii) calculate the molar mass, gas constant, density, specific heats and the changes in internal energy, enthalpy, entropy and Gibbs function of an ideal-gas mixture, including the entropy generated by mixing; (iv) define the psychrometric properties – dry bulb, wet bulb and dew point temperatures, specific and relative humidity, degree of saturation, enthalpy and specific volume of moist air – and compute them from the total pressure and steam-table data; (v) explain the adiabatic saturation process and the psychrometric chart, and read properties from the chart; (vi) analyse sensible heating and cooling, humidification, dehumidification, cooling with dehumidification, heating with humidification, evaporative cooling and adiabatic mixing, deriving the mass and energy balances and evaluating heat and moisture transfers; (vii) size the cooling coil, heating coil and humidifier of simple air-conditioning systems.

1. Introduction – mixtures of gases and moist air

Most gases met in engineering are mixtures. Air is a mixture of nitrogen, oxygen, argon, carbon dioxide and water vapour; the products of combustion are a mixture of CO_2 , H_2O , N_2 and O_2 ; natural gas, refinery gases and the gases in a cylinder of welding mixture are all mixtures. The first part of this unit shows how the properties of a mixture of ideal gases – molar mass, gas constant, density, specific heats, internal energy, enthalpy, entropy and Gibbs function – are found from the properties and proportions of its components, using Dalton's and Amagat's laws.

The second part treats the most important mixture of all: **moist air**, the mixture of dry air and water vapour that fills every room, factory and cooling tower. The water vapour is only a small fraction of the mass (never more than a few per cent) but it governs comfort, drying, condensation and the loads on air-conditioning plant. Its study is **psychrometry**, and its working tool is the **psychrometric chart**. The unit ends with the psychrometric processes – heating, cooling, humidifying, dehumidifying, evaporative cooling and mixing – that make up air-conditioning systems.

2. Composition of gas mixtures – mass and mole fractions, Dalton's and Amagat's laws

2.1 Mass fraction and mole fraction

Consider a mixture of k components of masses $m_1, m_2, \dots, m_k$ and mole numbers $n_1, n_2, \dots, n_k$, with $n_i = m_i/M_i$ (M_i = molar mass of component i). The total mass is $m = \sum m_i$ and the total number of kilomoles $n = \sum n_i$. The composition is described either by the **mass fraction** (gravimetric analysis) or by the **mole fraction** (molar or volumetric analysis):

$$\text{mass fraction } x_i = m_i/m, \quad \sum x_i = 1 \qquad \text{mole fraction } y_i = n_i/n, \quad \sum y_i = 1$$

The two are related through the molar masses: since $m_i = n_i M_i$, $x_i = y_i M_i/M$ and $y_i = (x_i/M_i)/\sum (x_j/M_j)$, where M is the molar mass of the mixture (Section 3.1). For ideal gases the mole fraction is also the *volume fraction* (Section 2.3), which is why a molar analysis is often called a volumetric analysis: air is 79 % N_2 and 21 % O_2 by volume (moles) but 76.7 % N_2 and 23.3 % O_2 by mass.

2.2 Dalton's law of partial pressures

The **partial pressure** p_i of a component is the pressure it would exert if it alone occupied the whole volume V of the mixture at the mixture temperature T (Figure 5.1, left, upper row). **Dalton's law:** *the pressure of*

a mixture of gases is the sum of the partial pressures of its components, $p = \sum p_i$, each component behaving as if the others were absent. For ideal gases $p_i V = n_i R_u T$ and $pV = n R_u T$, so

$$p = \sum p_i \quad p_i/p = n_i/n = y_i \quad p_i = y_i p \quad (\text{Dalton's law, ideal gases})$$

The partial pressure of a component is its mole fraction times the total pressure. The law is exact for ideal gases and a good approximation for real gases at low pressure; it is the basis of the whole of psychrometry, where the partial pressure of the water vapour in air is the quantity that matters.

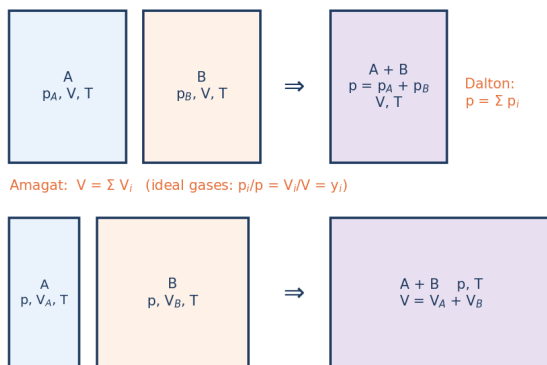
2.3 Amagat's law of partial volumes

The **partial volume** V_i of a component is the volume it would occupy if it alone existed at the pressure p and temperature T of the mixture (Figure 5.1, left, lower row). **Amagat's law (Leduc's law):** the volume of a mixture of gases is the sum of the partial volumes of its components, $V = \sum V_i$. For ideal gases $pV_i = n_i R_u T$, so

$$V = \sum V_i \quad V_i/V = n_i/n = y_i \quad (\text{Amagat's law, ideal gases})$$

For ideal gases Dalton's and Amagat's laws are identical in content – both state that $p_i/p = V_i/V = y_i$, so the mole fraction, the pressure fraction and the volume fraction are the same number. For real gases the two laws give slightly different results, Amagat's usually being the more accurate at high pressure because it accounts for the interactions between molecules at the actual mixture pressure.

Dalton's law (same V , T) and Amagat's law (same p , T)



Adiabatic mixing of two gases in a rigid tank (WE 5.2)

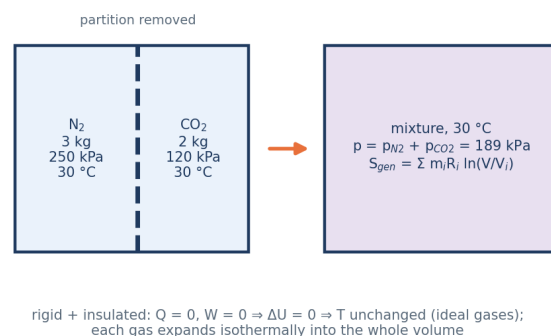


Figure 5.1 (Left) Dalton's law – components at the mixture volume and temperature, pressures add; Amagat's law – components at the mixture pressure and temperature, volumes add. (Right) Adiabatic mixing of two gases in a rigid insulated tank (Worked Example 5.2).

2.4 The Gibbs–Dalton law

Dalton's law is extended to all the extensive properties of an ideal-gas mixture by the **Gibbs–Dalton law**: the internal energy, enthalpy and entropy of a mixture of ideal gases are the sums of the internal energies, enthalpies and entropies that the components would have if each alone occupied the volume of the mixture at the mixture temperature – i.e. each component evaluated at its own partial pressure p_i and the common temperature T . This is the principle used in the next section.

3. Properties of ideal-gas mixtures

3.1 Molar mass and gas constant of the mixture

The mixture of total mass m and n kilomoles behaves as a single ideal gas of **apparent (equivalent) molar mass** $M = m/n$. Since $m = \sum n_i M_i$,

$$M = \sum y_i M_i = 1/\sum (x_i/M_i) \quad R = R_u/M = \sum x_i R_i \quad (\text{mixture gas constant, kJ/kg K})$$

The mixture obeys $pV = mRT = nR_u T$ with these values. For air, $M = 0.79 \times 28 + 0.21 \times 32 = 28.84$ kg/kmol (28.97 with argon and CO₂ included) and $R = 8.3145/28.97 = 0.287$ kJ/kg K.

3.2 Density of the mixture

The density is the total mass divided by the total volume; by Dalton's law each component fills the whole volume at its partial pressure, so

$$\rho = m/V = \sum \rho_i = \sum p_i/(R_i T) = p/(RT) \quad (\rho_i = \text{partial density of component } i)$$

3.3 Specific heats

By the Gibbs–Dalton law the internal energy and enthalpy of the mixture are $U = \sum m_i u_i = \sum n_i \bar{u}_i$ and $H = \sum m_i h_i = \sum n_i \bar{h}_i$. Differentiating with respect to temperature gives the specific heats of the mixture as the mass-fraction-weighted (or, on a molar basis, the mole-fraction-weighted) averages of those of the components:

$$c_v = \sum x_i c_{v,i} \quad c_p = \sum x_i c_{p,i} \quad \bar{c}_v = \sum y_i \bar{c}_{v,i} \quad \bar{c}_p = \sum y_i \bar{c}_{p,i} \quad c_p - c_v = R, \quad \gamma = c_p/c_v$$

3.4 Changes in internal energy and enthalpy

For a process of the mixture from temperature T_1 to T_2 (ideal gases: u and h depend on T only):

$$\Delta U = \sum m_i c_{v,i} (T_2 - T_1) = m c_v (T_2 - T_1) \quad \Delta H = \sum m_i c_{p,i} (T_2 - T_1) = m c_p (T_2 - T_1)$$

The changes are the sums of the changes of the components, and are the same whether the components are mixed or separate.

3.5 Change in entropy and the entropy of mixing

Entropy depends on pressure as well as temperature, and by the Gibbs–Dalton law each component must be evaluated at its **partial pressure**. For a process $1 \rightarrow 2$ of the mixture,

$$\Delta S = \sum m_i [c_{p,i} \ln(T_2/T_1) - R_i \ln(p_{i,2}/p_{i,1})] = \sum n_i [\bar{c}_{p,i} \ln(T_2/T_1) - R_u \ln(p_{i,2}/p_{i,1})]$$

Entropy of mixing. When gases initially separate, each at pressure p and temperature T , are mixed adiabatically to the same p and T (a partition removed between compartments of an insulated vessel, Figure 5.1, right), the temperature is unchanged ($\Delta U = 0$ for ideal gases with $Q = W = 0$), but each component now exists at its partial pressure $p_i = y_i p < p$, so its entropy increases:

$$\Delta S_{\text{mix}} = -R_u \sum n_i \ln(p_i/p) = -R_u \sum n_i \ln y_i = -n R_u \sum y_i \ln y_i > 0 \quad (\text{equivalently } \Delta S_{\text{mix}} = \sum m_i R_i \ln(V/V_i))$$

Since every $y_i < 1$ the entropy of mixing is always positive: **mixing of different gases is irreversible**, and the entropy generated is independent of the nature of the gases – it depends only on the mole numbers. The Gibbs paradox notes that the same formula would apply to "mixing" two samples of the *same* gas, which produces no change of state; the formula therefore applies only to distinguishable gases. When the components are initially at different pressures (Worked Example 5.2) the general form $\sum m_i R_i \ln(V/V_i)$ is used, each gas expanding isothermally from its initial volume V_i to the whole volume V .

3.6 Gibbs function of a mixture

The Gibbs function $g = h - Ts$ of an ideal-gas mixture is $G = \sum n_i \bar{g}_i$ with each component evaluated at its partial pressure: $\bar{g}_i(T, p_i) = \bar{g}_i(T, p) + R_u T \ln y_i$. Hence the Gibbs function of the mixture is less than the sum for the unmixed gases by $n R_u T \sum y_i \ln y_i (= -T \Delta S_{\text{mix}})$, which is the "free energy" lost in mixing – the maximum work that could have been obtained by mixing the gases reversibly (through semi-permeable membranes), and the minimum work needed to separate them again, e.g. in an air-separation plant. The Gibbs function is the property that governs chemical equilibrium in reacting mixtures.

KEY POINT For an ideal-gas mixture: u , h , c_v and c_p are mass-weighted averages and depend on T only; entropy must be evaluated with each component at its *partial* pressure; the temperature of a mixture is common to all components, but each component "sees" only its own partial pressure.

3.7 Worked examples on gas mixtures

WORKED EXAMPLE 5.1 — Composition, gas constant and partial pressures of a gas mixture

Problem. A gas mixture has the composition by mass: nitrogen 55 %, oxygen 30 %, carbon dioxide 15 %. Find (i) the mole fraction of each constituent, (ii) the molar mass and gas constant of the mixture, (iii) the partial pressure of each constituent when the mixture is at 250 kPa and 320 K and (iv) the volume of 4 kg of the mixture at these conditions. Molar masses: N_2 28, O_2 32, CO_2 44 kg/kmol.

Given: $x_{N_2} = 0.55$, $x_{O_2} = 0.30$, $x_{CO_2} = 0.15$; $p = 250$ kPa; $T = 320$ K; $m = 4$ kg; $R_u = 8.3145$ kJ/kmol K.

Formula: $n_i = x_i/M_i$ (per kg of mixture); $y_i = n_i/\sum n_i$; $M = 1/(\sum (x_i/M_i))$; $R = R_u/M$; $p_i = y_i p$; $V = mRT/p$.

Solution.

(i) Per kg of mixture: $n_{N_2} = 0.55/28 = 0.019643$ kmol; $n_{O_2} = 0.30/32 = 0.009375$ kmol; $n_{CO_2} = 0.15/44 = 0.003409$ kmol; total $n = 0.032427$ kmol.

$y_{N_2} = 0.019643/0.032427 = 0.606$; $y_{O_2} = 0.009375/0.032427 = 0.289$; $y_{CO_2} = 0.003409/0.032427 = 0.105$ (sum 1.000 ✓).

(ii) $M = 1/0.032427 = 30.84$ kg/kmol (check: $\sum y_i M_i = 0.606 \times 28 + 0.289 \times 32 + 0.105 \times 44 = 30.84$ ✓); $R = 8.3145/30.84 = 0.2696$ kJ/kg K.

(iii) $p_{N_2} = 0.606 \times 250 = 151.4$ kPa; $p_{O_2} = 0.289 \times 250 = 72.3$ kPa; $p_{CO_2} = 0.105 \times 250 = 26.3$ kPa (sum 250 kPa ✓).

(iv) $V = 4 \times 0.2696 \times 320/250 = 1.380$ m³.

ANSWER $y_{N_2} = 0.606$, $y_{O_2} = 0.289$, $y_{CO_2} = 0.105$; $M = 30.84$ kg/kmol, $R = 0.2696$ kJ/kg K; partial pressures 151.4, 72.3 and 26.3 kPa; $V = 1.38$ m³. Note that the heavy CO_2 is 15 % of the mass but only 10.5 % of the moles (and of the volume), while the light nitrogen is 55 % of the mass but 60.6 % of the moles.

WORKED EXAMPLE 5.2 — Adiabatic mixing of two gases in a rigid tank – entropy generation

Problem. A rigid insulated tank is divided by a partition into two compartments; one holds 3 kg of nitrogen at 250 kPa and 30 °C, the other 2 kg of carbon dioxide at 120 kPa and 30 °C. The partition is removed and the gases mix. Find (i) the volume of the tank, (ii) the final pressure, (iii) the partial pressure of each gas after mixing and (iv) the entropy generated. $R_{N_2} = 0.2968$ kJ/kg K, $R_{CO_2} = 0.1889$ kJ/kg K.

Given: $m_N = 3$ kg, $p_N = 250$ kPa; $m_C = 2$ kg, $p_C = 120$ kPa; $T = 303.15$ K (both); insulated rigid tank.

Formula: $V_i = m_i R_i T / p_i$; $V = \sum V_i$; after mixing (T unchanged, since $Q = W = 0$ and $\Delta U = 0$ for ideal gases) $p_{i,2} = m_i R_i T / V$; $p = \sum p_{i,2}$; $S_{\text{gen}} = \sum m_i R_i \ln(V/V_i) = \sum m_i R_i \ln(p_{i,1}/p_{i,2})$ (isothermal expansion of each gas).

Solution.

(i) $V_N = 3 \times 0.2968 \times 303.15/250 = 1.080 \text{ m}^3$; $V_C = 2 \times 0.1889 \times 303.15/120 = 0.954 \text{ m}^3$; $V = 1.080 + 0.954 = 2.034 \text{ m}^3$.

(ii), (iii) $p_{N,2} = 3 \times 0.2968 \times 303.15/2.034 = 132.7 \text{ kPa}$; $p_{C,2} = 2 \times 0.1889 \times 303.15/2.034 = 56.3 \text{ kPa}$; $p = 132.7 + 56.3 = 189.0 \text{ kPa}$.

(iv) The temperature stays at 30°C , so each gas simply expands isothermally into the whole tank:
 $S_{\text{gen}} = 3 \times 0.2968 \times \ln(2.034/1.080) + 2 \times 0.1889 \times \ln(2.034/0.954) = 0.8904 \times 0.6333 + 0.3778 \times 0.7569 = 0.564 + 0.286 = 0.850 \text{ kJ/K}$.

ANSWER (i) $V = 2.034 \text{ m}^3$; (ii) $p = 189.0 \text{ kPa}$; (iii) $p_{N_2} = 132.7 \text{ kPa}$, $p_{\text{CO}_2} = 56.3 \text{ kPa}$; (iv) $S_{\text{gen}} = 0.850 \text{ kJ/K}$. The entropy rises although no heat is transferred – the mixing (and the expansion of each gas through a finite pressure difference) is irreversible; with surroundings at 30°C the work potential destroyed is $T_0 S_{\text{gen}} = 303.15 \times 0.850 = 258 \text{ kJ}$.

4. Psychrometric properties of moist air and the psychrometric chart

4.1 Moist air as an ideal-gas mixture

Psychrometry is the study of the properties of moist air – the mixture of dry air and water vapour – and of the processes it undergoes; its applications include the design of air-conditioning, ventilation and refrigeration systems, cooling towers, drying of food, timber and textiles, and the study of human comfort.

Dry air is treated as a single ideal gas with $M = 28.97 \text{ kg/kmol}$, $R_a = 0.287 \text{ kJ/kg K}$ and $c_{pa} = 1.005 \text{ kJ/kg K}$. The **water vapour** in atmospheric air is at a very low partial pressure (0.5 to 6 kPa) and is superheated at ordinary temperatures, so it too behaves as an ideal gas with $R_v = 0.4615 \text{ kJ/kg K}$; its enthalpy depends on temperature alone and is closely approximated by that of saturated vapour at the same temperature, $h_v \approx h_g(t) \approx 2501 + 1.88t \text{ kJ/kg}$ (t in $^\circ\text{C}$). By Dalton's law the total (barometric) pressure is

$$p = p_a + p_v \quad (p_a = \text{partial pressure of dry air, } p_v = \text{partial pressure of water vapour})$$

Because the mass of vapour changes during humidification and dehumidification while the mass of dry air does not, all psychrometric properties are expressed **per kilogram of dry air**.

4.2 Definitions

- **Dry bulb temperature (DBT), t :** the temperature of the moist air measured by an ordinary thermometer whose bulb is dry and shielded from radiation.
- **Wet bulb temperature (WBT), t_{wb} :** the steady temperature registered by a thermometer whose bulb is covered with a wick kept wet with water and exposed to a rapid stream (about 3 m/s or more) of the air. Water evaporates from the wick into the unsaturated air, cooling the bulb until the heat gained by convection from the air balances the latent heat of evaporation; the WBT lies between the DBT and the dew point, and equals the DBT only when the air is saturated. The difference $t - t_{wb}$ is the *wet bulb depression*, a measure of the dryness of the air.
- **Dew point temperature (DPT), t_{dp} :** the temperature at which the water vapour in the air begins to condense when the air is cooled at constant pressure (and constant specific humidity); it is the saturation temperature corresponding to the partial pressure of the vapour, $t_{dp} = T_{\text{sat}}(p_v)$. For saturated air $\text{DBT} = \text{WBT} = \text{DPT}$.

- **Specific humidity (humidity ratio, moisture content), ω :** the mass of water vapour per unit mass of dry air. With the ideal-gas equation for both components at the same V and T ,

$$\omega = m_v/m_a = (p_v V/R_v T)/(p_a V/R_a T) = 0.622 p_v/p_a = 0.622 p_v/(p - p_v) \quad [\text{kg vapour/kg dry air}]$$

($0.622 = R_a/R_v = 18.015/28.97$). The specific humidity depends on the partial pressure of the vapour and on the total pressure; it does not change in sensible heating or cooling. Its value ranges from zero (dry air) to about 0.03 kg/kg at 33 °C saturated.

- **Saturated air:** air holding the maximum amount of vapour possible at its temperature – the vapour is saturated, $p_v = p_s = p_{\text{sat}}(t)$, and $\omega_s = 0.622 p_s/(p - p_s)$. Any further vapour, or any cooling, causes condensation (fog, dew).
- **Relative humidity, ϕ :** the ratio of the partial pressure of the vapour in the air to the saturation pressure of water at the same temperature (equivalently the ratio of the mass of vapour present to the mass the air could hold when saturated at the same temperature and total pressure, or of the vapour densities):

$$\phi = p_v/p_s = \rho_v/\rho_s = m_v/m_s \quad (0 \leq \phi \leq 1; \phi = 1 \text{ for saturated air})$$

Relative humidity indicates how near the air is to saturation and hence how readily it will take up moisture (drying, sweating); for a fixed ω it *falls* as the air is heated, because p_s rises with temperature. Specific humidity measures the *quantity* of vapour; relative humidity measures its *nearness to saturation* – the two are related by $\phi = \omega p/[(0.622 + \omega)p_s]$.

- **Degree of saturation (percentage humidity), μ :** the ratio of the actual specific humidity to that of saturated air at the same temperature and pressure:

$$\mu = \omega/\omega_s = \phi (p - p_s)/(p - p_v) \quad (\mu \leq \phi; \text{nearly equal at low temperatures})$$

- **Enthalpy of moist air, h :** the enthalpy of the dry air plus that of the vapour, per kg of dry air, with the datum of 0 °C for dry air and saturated liquid at 0 °C for water:

$$h = h_a + \omega h_v = c_{pa}t + \omega h_g(t) \approx 1.005t + \omega(2501 + 1.88t) \quad [\text{kJ/kg dry air}]$$

The first term is the *sensible heat* (it changes with temperature) and the second the *latent heat* carried by the vapour (it changes with moisture content); processes that change only t are sensible processes, those that change ω involve latent heat.

- **Specific volume, v :** the volume of the mixture per kg of dry air, $v = V/m_a = R_a T/p_a = R_a T/(p - p_v)$ [$\text{m}^3/\text{kg dry air}$]; the density of the dry air in the mixture is $\rho_a = p_a/(R_a T) = 1/v$ and that of the vapour $\rho_v = p_v/(R_v T)$, the density of the moist air being $\rho_a + \rho_v$.

4.3 Adiabatic saturation and the wet bulb temperature

In an **adiabatic saturator** (Figure 5.3, left) unsaturated air at t_1 , ω_1 flows steadily through a long insulated chamber over a pool of water; water evaporates into the air, the latent heat being supplied by the air itself, so the air cools and its humidity rises until it leaves saturated at a lower temperature t_2 . Make-up water is supplied at t_2 to keep the pool level constant. The exit temperature t_2 , which depends only on the inlet state and the pressure, is the **adiabatic saturation temperature (thermodynamic wet bulb temperature)** of the entering air. Balancing mass and energy per kg of dry air (no heat transfer, no work):

$$\text{water added} = \omega_2 - \omega_1; \quad h_1 + (\omega_2 - \omega_1)h_{f2} = h_2,$$

$$\text{i.e. } c_{pa}t_1 + \omega_1 h_{g1} + (\omega_2 - \omega_1)h_{f2} = c_{pa}t_2 + \omega_2 h_{g2}. \text{ Solving for } \omega_1,$$

$$\omega_1 = [c_{pa}(t_2 - t_1) + \omega_2 h_{fg2}]/(h_{g1} - h_{f2}) \quad \text{with } \omega_2 = 0.622 p_{s2}/(p - p_{s2})$$

Thus a measurement of t_1 , t_2 and p gives ω_1 , and hence p_{v1} , ϕ_1 and t_{dp1} – the adiabatic saturator is a humidity-measuring instrument (Worked Example 5.5). Since $(\omega_2 - \omega_1)h_{f2}$ is small, the process is very nearly one of **constant enthalpy**, and the lines of constant adiabatic saturation temperature on the psychrometric chart almost coincide with the lines of constant enthalpy.

For air–water vapour mixtures at ordinary conditions the **wet bulb temperature** read by a wet-bulb thermometer or sling psychrometer is very nearly equal to the adiabatic saturation temperature (the ratio of the heat- and mass-transfer coefficients, the Lewis number, is close to unity for this mixture), so in practice t_{wb} is used in the equation above and the constant-WBT lines are drawn on the chart. The psychrometer (dry- and wet-bulb thermometers) is therefore the standard instrument for measuring the humidity of air.

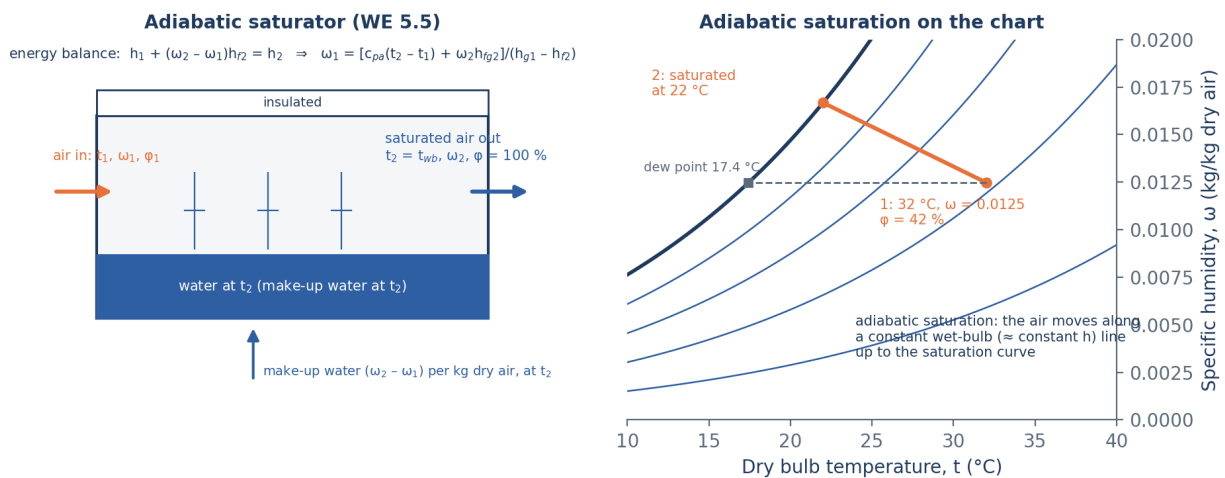


Figure 5.3 (Left) The adiabatic saturator. (Right) The adiabatic saturation process on the psychrometric chart – the air follows a constant wet-bulb (nearly constant-enthalpy) line up to the saturation curve; the dew point of the inlet air is found by moving horizontally to the saturation curve.

4.4 The psychrometric chart

The properties of moist air at a fixed total pressure (usually 101.325 kPa, sea level) are plotted on the **psychrometric chart** (Figure 5.2), which allows any state to be fixed from two properties and the others to be read without calculation. The chart has:

- **Horizontal axis:** dry bulb temperature t (°C); vertical lines are constant-DBT lines.
- **Vertical axis (on the right):** specific humidity ω (kg/kg dry air), sometimes also the vapour pressure p_v , to which ω is nearly proportional; horizontal lines are constant- ω lines.
- **Saturation curve** ($\phi = 100\%$): the upper boundary of the chart, giving $\omega_s(t)$; the DPT of any state is read where the horizontal through the state meets this curve.
- **Constant relative humidity curves** (10 %, 20 % ... 90 %) fanning out below the saturation curve.
- **Constant wet bulb temperature lines**, sloping down to the right from the saturation curve (a WBT line meets the saturation curve at $t = t_{wb}$).
- **Constant enthalpy lines** (kJ/kg dry air), almost parallel to the WBT lines, often drawn with an enthalpy scale above the saturation curve.
- **Constant specific volume lines** (m^3/kg dry air), steeply sloping down to the right.
- Some charts also carry a sensible heat factor protractor and correction scales for pressure.

To use the chart, fix the state by any two of DBT, WBT, DPT, ω , ϕ , h and v , then read the rest (Worked Examples 5.6 to 5.8). Processes are straight lines or simple curves on the chart, which makes it the everyday tool of the air-conditioning engineer.

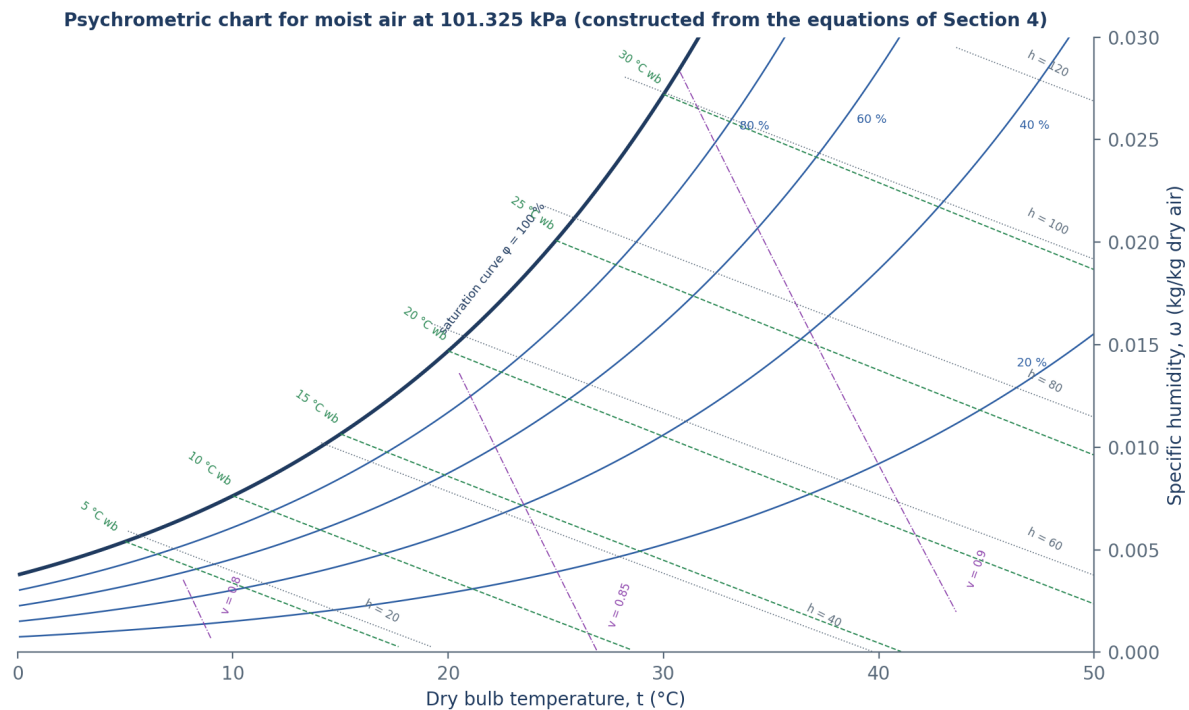


Figure 5.2 The psychrometric chart for moist air at 101.325 kPa, constructed from the equations of this section: saturation curve (navy), constant relative humidity (blue), constant wet bulb temperature (green dashed), constant enthalpy in kJ/kg dry air (grey dotted) and constant specific volume in m³/kg dry air (purple dash-dot).

KEY POINT Property calculation always starts from the vapour pressure: from ϕ ($p_v = \phi p_s$), from the dew point ($p_v = p_{\text{sat}}(t_{\text{dp}})$) or from the wet bulb (adiabatic-saturation equation $\rightarrow \omega \rightarrow p_v$). Then $\omega = 0.622p_v/(p - p_v)$, $h = 1.005t + \omega(2501 + 1.88t)$, $v = 0.287(t + 273.15)/(p - p_v)$. The steam tables supply p_s , h_g , h_f and h_{fg} . Work per kg of dry air throughout.

4.5 Worked examples on psychrometric properties

WORKED EXAMPLE 5.3 — Complete property calculation from dry and wet bulb temperatures

Problem. Atmospheric air at 1.0132 bar has a dry bulb temperature of 34 °C and a wet bulb temperature of 25 °C. Calculate (i) the partial pressure of the water vapour, (ii) the specific humidity, (iii) the dew point temperature, (iv) the relative humidity, (v) the degree of saturation, (vi) the density of the dry air, (vii) the density of the water vapour and (viii) the enthalpy of the mixture per kg of dry air.

Given: $p = 101.32 \text{ kPa}$; $t = 34 \text{ °C}$; $t_{\text{wb}} = 25 \text{ °C}$. Steam tables: $p_s(25 \text{ °C}) = 3.170 \text{ kPa}$, $h_f(25 \text{ °C}) = 104.8 \text{ kJ/kg}$, $h_{fg}(25 \text{ °C}) = 2441.7 \text{ kJ/kg}$; $p_s(34 \text{ °C}) = 5.325 \text{ kPa}$, $h_g(34 \text{ °C}) = 2562.8 \text{ kJ/kg}$.

Formula: $\omega_{\text{wb}} = 0.622p_{s,\text{wb}}/(p - p_{s,\text{wb}})$; $\omega = [c_{p,a}(t_{\text{wb}} - t) + \omega_{\text{wb}}h_{f,g,\text{wb}}]/(h_g - h_{f,\text{wb}})$; $p_v = \omega p/(0.622 + \omega)$; $t_{\text{dp}} = T_{\text{sat}}(p_v)$; $\phi = p_v/p_s$; $\mu = \omega/\omega_s$; $\rho_a = (p - p_v)/(R_a T)$; $\rho_v = p_v/(R_v T)$; $h = 1.005t + \omega(2501 + 1.88t)$.

Solution.

Saturated humidity at the wet bulb temperature: $\omega_{\text{wb}} = 0.622 \times 3.170/(101.32 - 3.170) = 1.972/98.15 = 0.02009 \text{ kg/kg}$.

(ii) $\omega = [1.005 \times (25 - 34) + 0.02009 \times 2441.7]/(2562.8 - 104.8) = (-9.05 + 49.05)/2458.0 = 40.00/2458.0 = 0.01628 \text{ kg/kg dry air}$.

(i) $p_v = 0.01628 \times 101.32/(0.622 + 0.01628) = 1.649/0.6383 = 2.584 \text{ kPa}$.

(iii) $t_{\text{dp}} = \text{saturation temperature at } 2.584 \text{ kPa} = 21.6 \text{ °C}$ (tables: 2.487 kPa at 21 °C, 2.645 kPa at 22 °C).

(iv) $\phi = 2.584/5.325 = 0.485 = 48.5 \%$.

(v) $\omega_s = 0.622 \times 5.325/(101.32 - 5.325) = 3.312/95.995 = 0.03450$; $\mu = 0.01628/0.03450 = 0.472$.

(vi) $p_a = (101.32 - 2.584)/(0.287 \times 307.15) = 98.74/88.15 = 1.120 \text{ kg/m}^3$.
 (vii) $p_v = 2.584/(0.4615 \times 307.15) = 2.584/141.75 = 0.01823 \text{ kg/m}^3$.
 (viii) $h = 1.005 \times 34 + 0.01628 \times (2501 + 1.88 \times 34) = 34.17 + 0.01628 \times 2564.9 = 34.17 + 41.76 = 75.9 \text{ kJ/kg dry air}$.

ANSWER $p_v = 2.584 \text{ kPa}$; $\omega = 0.01628 \text{ kg/kg}$; $t_{dp} = 21.6 \text{ }^\circ\text{C}$; $\phi = 48.5 \%$; $\mu = 0.472$; $\rho_a = 1.120 \text{ kg/m}^3$; $\rho_v = 0.0182 \text{ kg/m}^3$ ($\rho_v/\rho_a = \omega$); $h = 75.9 \text{ kJ/kg dry air}$. A wet bulb depression of 9 K corresponds to a relative humidity of only about 48 % – the air is far from saturated and evaporative cooling would be effective.

WORKED EXAMPLE 5.4 — Moist air in a room – partial pressure, humidity, enthalpy and masses

Problem. A room 5 m × 4 m × 3 m contains air at 28 °C and 100 kPa with a relative humidity of 60 %. Find (i) the partial pressure of the dry air, (ii) the specific humidity, (iii) the enthalpy per kg of dry air and (iv) the masses of dry air and of water vapour in the room.

Given: $V = 60 \text{ m}^3$; $t = 28 \text{ }^\circ\text{C}$ (301.15 K); $p = 100 \text{ kPa}$; $\phi = 0.60$; $p_s(28 \text{ }^\circ\text{C}) = 3.783 \text{ kPa}$; $R_a = 0.287$, $R_v = 0.4615 \text{ kJ/kg K}$.

Formula: $p_v = \phi p_s$; $p_a = p - p_v$; $\omega = 0.622 p_v/p_a$; $h = 1.005t + \omega(2501 + 1.88t)$; $m_a = p_a V/(R_a T)$; $m_v = \omega m_a = p_v V/(R_v T)$.

Solution.

$$p_v = 0.60 \times 3.783 = 2.270 \text{ kPa}.$$

$$(i) p_a = 100 - 2.270 = 97.73 \text{ kPa}.$$

$$(ii) \omega = 0.622 \times 2.270/97.73 = 0.01445 \text{ kg/kg dry air}.$$

$$(iii) h = 1.005 \times 28 + 0.01445 \times (2501 + 1.88 \times 28) = 28.14 + 0.01445 \times 2553.6 = 28.14 + 36.90 = 65.0 \text{ kJ/kg dry air}.$$

$$(iv) m_a = 97.73 \times 60/(0.287 \times 301.15) = 5863.8/86.43 = 67.8 \text{ kg}; m_v = 0.01445 \times 67.8 = 0.98 \text{ kg}$$

(check: $p_v V/(R_v T) = 2.270 \times 60/(0.4615 \times 301.15) = 136.2/139.0 = 0.98 \text{ kg } \checkmark$).

ANSWER $p_a = 97.73 \text{ kPa}$; $\omega = 0.01445 \text{ kg/kg}$; $h = 65.0 \text{ kJ/kg dry air}$; $m_a = 67.8 \text{ kg}$, $m_v = 0.98 \text{ kg}$. The vapour is only 1.4 % of the mass of the air in the room, yet it carries 57 % of the enthalpy above the datum.

WORKED EXAMPLE 5.5 — Adiabatic saturator – humidity of the inlet air

Problem. An air–water vapour mixture enters an adiabatic saturator at 32 °C and leaves saturated at 22 °C; the pressure is constant at 101.325 kPa. Determine the humidity ratio, the relative humidity and the dew point temperature of the inlet air and show the process on the psychrometric chart.

Given: $t_1 = 32 \text{ }^\circ\text{C}$; $t_2 = 22 \text{ }^\circ\text{C}$; $p = 101.325 \text{ kPa}$. Tables: $p_s(22 \text{ }^\circ\text{C}) = 2.645 \text{ kPa}$, $h_f(22) = 92.3 \text{ kJ/kg}$, $h_{fg}(22) = 2448.8 \text{ kJ/kg}$; $p_s(32 \text{ }^\circ\text{C}) = 4.759 \text{ kPa}$, $h_g(32) = 2559.2 \text{ kJ/kg}$.

Formula: $\omega_2 = 0.622 p_{s2}/(p - p_{s2})$; $\omega_1 = [c_{pa}(t_2 - t_1) + \omega_2 h_{fg2}]/(h_{g1} - h_{f2})$; $p_{v1} = \omega_1 p/(0.622 + \omega_1)$; $\phi_1 = p_{v1}/p_{s1}$; $t_{dp} = T_{sat}(p_{v1})$.

Solution.

$$\omega_2 = 0.622 \times 2.645/(101.325 - 2.645) = 1.645/98.68 = 0.01667 \text{ kg/kg}.$$

$$\omega_1 = [1.005 \times (22 - 32) + 0.01667 \times 2448.8]/(2559.2 - 92.3) = (-10.05 + 40.82)/2466.9 = 30.77/2466.9 = 0.01248 \text{ kg/kg dry air}.$$

$$p_{v1} = 0.01248 \times 101.325/(0.622 + 0.01248) = 1.2645/0.6345 = 1.993 \text{ kPa}.$$

$$\phi_1 = 1.993/4.759 = 0.419 = 41.9 \%$$

$$t_{dp} = T_{sat}(1.993 \text{ kPa}) = 17.4 \text{ }^\circ\text{C} \text{ (tables: 1.938 kPa at } 17 \text{ }^\circ\text{C, 2.065 kPa at } 18 \text{ }^\circ\text{C)}.$$

ANSWER $\omega_1 = 0.0125 \text{ kg/kg dry air}$; $\phi_1 = 42 \%$; $t_{dp} = 17.4 \text{ }^\circ\text{C}$. On the chart (Figure 5.3, right) the process runs from state 1 ($32 \text{ }^\circ\text{C}$, $\omega = 0.0125$) up the $22 \text{ }^\circ\text{C}$ wet-bulb line to the saturation curve at $22 \text{ }^\circ\text{C}$; the make-up water is $0.01667 - 0.01248 = 0.0042 \text{ kg per kg of dry air}$.

WORKED EXAMPLE 5.6 — Properties from the chart – dry bulb and wet bulb given

Problem. Atmospheric air at 1.01325 bar has a dry bulb temperature of $32 \text{ }^\circ\text{C}$ and a wet bulb temperature of $24 \text{ }^\circ\text{C}$. Using the psychrometric chart find (i) the specific humidity, (ii) the relative humidity, (iii) the dew point temperature, (iv) the enthalpy and (v) the specific volume of the moist air.

Given: $t = 32 \text{ }^\circ\text{C}$, $t_{wb} = 24 \text{ }^\circ\text{C}$, $p = 101.325 \text{ kPa}$ (the standard chart).

Formula: Locate the intersection of the $32 \text{ }^\circ\text{C}$ vertical with the $24 \text{ }^\circ\text{C}$ wet-bulb line; read ω on the right-hand scale, ϕ from the curves, t_{dp} by moving horizontally to the saturation curve, h along the enthalpy line, v from the volume lines.

Solution.

(i) $\omega = 0.0155 \text{ kg/kg dry air}$ (calculation: 0.01550).

(ii) $\phi = 52 \%$ (calculation: $p_v = 2.464 \text{ kPa}$, $p_s = 4.759 \text{ kPa}$, $\phi = 0.518$).

(iii) $t_{dp} = 20.8 \text{ }^\circ\text{C}$.

(iv) $h = 71.9 \text{ kJ/kg dry air}$ (calculation: $1.005 \times 32 + 0.0155 \times 2561.2 = 32.16 + 39.70 = 71.9$).

(v) $v = 0.886 \text{ m}^3/\text{kg dry air}$ (calculation: $0.287 \times 305.15/(101.325 - 2.464) = 87.58/98.86 = 0.886$).

ANSWER $\omega = 0.0155 \text{ kg/kg}$; $\phi = 52 \%$; $t_{dp} = 20.8 \text{ }^\circ\text{C}$; $h = 71.9 \text{ kJ/kg dry air}$; $v = 0.886 \text{ m}^3/\text{kg dry air}$. The chart values agree with the calculated ones to within the reading accuracy of the chart (about $\pm 0.0003 \text{ kg/kg}$, $\pm 2 \%$ in ϕ , $\pm 0.5 \text{ kJ/kg}$).

WORKED EXAMPLE 5.7 — Chart properties and cooling to below the dew point

Problem. Moist air at 1.01325 bar has a dry bulb temperature of $36 \text{ }^\circ\text{C}$ and a relative humidity of 45 %. From the chart find the specific humidity, dew point, wet bulb temperature, enthalpy and specific volume. The air is then cooled at constant pressure to $12 \text{ }^\circ\text{C}$; show on the chart whether condensation occurs and, if so, find the moisture removed per kg of dry air.

Given: $t_1 = 36 \text{ }^\circ\text{C}$, $\phi_1 = 0.45$, $p = 101.325 \text{ kPa}$; final $t_2 = 12 \text{ }^\circ\text{C}$. (Tables for the check: $p_s(36) = 5.947 \text{ kPa}$; $p_s(12) = 1.403 \text{ kPa}$.)

Formula: Chart readings at the intersection of the $36 \text{ }^\circ\text{C}$ vertical and the 45 % curve; cooling at constant ω is a horizontal line to the left – condensation begins where it meets the saturation curve (the dew point); below the dew point the state follows the saturation curve and ω falls to $\omega_s(t_2)$; moisture removed = $\omega_1 - \omega_2$.

Solution.

State 1 from the chart: $\omega_1 = 0.0169 \text{ kg/kg dry air}$ (calc. 0.01688); $t_{dp1} = 22.2 \text{ }^\circ\text{C}$; $t_{wb1} = 25.9 \text{ }^\circ\text{C}$; $h_1 = 79.5 \text{ kJ/kg dry air}$; $v_1 = 0.899 \text{ m}^3/\text{kg dry air}$.

Cooling: the horizontal line from state 1 meets the saturation curve at $22.2 \text{ }^\circ\text{C}$. Since the final temperature $12 \text{ }^\circ\text{C}$ is below the dew point, **condensation occurs**; from $22.2 \text{ }^\circ\text{C}$ down to $12 \text{ }^\circ\text{C}$ the state slides down the saturation curve.

State 2 (saturated at $12 \text{ }^\circ\text{C}$): $\omega_2 = 0.622 \times 1.403/(101.325 - 1.403) = 0.00873 \text{ kg/kg dry air}$; $h_2 = 34.1 \text{ kJ/kg dry air}$.

Moisture removed = $0.0169 - 0.0087 = 0.0082 \text{ kg per kg of dry air}$.

ANSWER $\omega = 0.0169 \text{ kg/kg}$, $t_{dp} = 22.2 \text{ }^\circ\text{C}$, $t_{wb} = 25.9 \text{ }^\circ\text{C}$, $h = 79.5 \text{ kJ/kg}$, $v = 0.899 \text{ m}^3/\text{kg}$ (all per kg dry air); on cooling to $12 \text{ }^\circ\text{C}$ about 8.2 g of water condenses per kg of dry air – almost half the moisture – and the heat removed is $79.5 - 34.1 \approx 45 \text{ kJ/kg}$ dry air (less the small enthalpy of the condensate). This is the cooling-and-dehumidifying process of a summer air conditioner.

WORKED EXAMPLE 5.8 — Chart properties from dry bulb and dew point, with verification

Problem. Moist air at 1.01325 bar has a dry bulb temperature of $30 \text{ }^\circ\text{C}$ and a dew point temperature of $18 \text{ }^\circ\text{C}$. From the chart find (i) the specific humidity, (ii) the relative humidity, (iii) the wet bulb temperature, (iv) the enthalpy and (v) the specific volume. Verify the specific and relative humidity by calculation using the saturation pressures at $18 \text{ }^\circ\text{C}$ and $30 \text{ }^\circ\text{C}$ from the steam tables.

Given: $t = 30 \text{ }^\circ\text{C}$, $t_{dp} = 18 \text{ }^\circ\text{C}$, $p = 101.325 \text{ kPa}$; $p_s(18 \text{ }^\circ\text{C}) = 2.065 \text{ kPa}$; $p_s(30 \text{ }^\circ\text{C}) = 4.247 \text{ kPa}$.

Formula: Chart: the state is the intersection of the $30 \text{ }^\circ\text{C}$ vertical with the horizontal through the saturation curve at $18 \text{ }^\circ\text{C}$. Calculation: $p_v = p_s(t_{dp})$; $\omega = 0.622 p_v / (p - p_v)$; $\phi = p_v / p_s(t)$.

Solution.

From the chart: (i) $\omega = 0.0129 \text{ kg/kg}$ dry air; (ii) $\phi = 49 \%$; (iii) $t_{wb} = 21.7 \text{ }^\circ\text{C}$; (iv) $h = 63.2 \text{ kJ/kg}$ dry air; (v) $v = 0.877 \text{ m}^3/\text{kg}$ dry air.

Verification: $p_v = p_s(18 \text{ }^\circ\text{C}) = 2.065 \text{ kPa}$; $\omega = 0.622 \times 2.065 / (101.325 - 2.065) = 1.284 / 99.26 = 0.01294 \text{ kg/kg}$ ✓; $\phi = 2.065 / 4.247 = 0.486 = 48.6 \%$ ✓. (Also $h = 1.005 \times 30 + 0.01294 \times 2557.4 = 30.15 + 33.09 = 63.2 \text{ kJ/kg}$ ✓ and $v = 0.287 \times 303.15 / 99.26 = 0.877 \text{ m}^3/\text{kg}$ ✓.)

ANSWER $\omega = 0.0129 \text{ kg/kg}$; $\phi = 49 \%$; $t_{wb} = 21.7 \text{ }^\circ\text{C}$; $h = 63.2 \text{ kJ/kg}$ dry air; $v = 0.877 \text{ m}^3/\text{kg}$ dry air; the calculated $\omega = 0.01294$ and $\phi = 48.6 \%$ confirm the chart readings.

5. Psychrometric processes

Air-conditioning consists of changing the temperature and moisture content of air by one or more of the processes below, each of which is a straight line or a simple path on the psychrometric chart (Figure 5.4, left). Every process is analysed with the steady-flow mass balance for water and the energy balance, written per kg of dry air (the dry-air flow $\dot{m}_a$ is the same at inlet and exit):

$$\text{water: } \dot{m}_a \omega_1 + \dot{m}_w = \dot{m}_a \omega_2 \quad \text{energy: } \dot{m}_a h_1 + \dot{Q} + \dot{m}_w h_w = \dot{m}_a h_2$$

where $\dot{m}_w$ is the water or steam added (negative if condensate is removed) with enthalpy h_w , and $\dot{Q}$ the heat added.

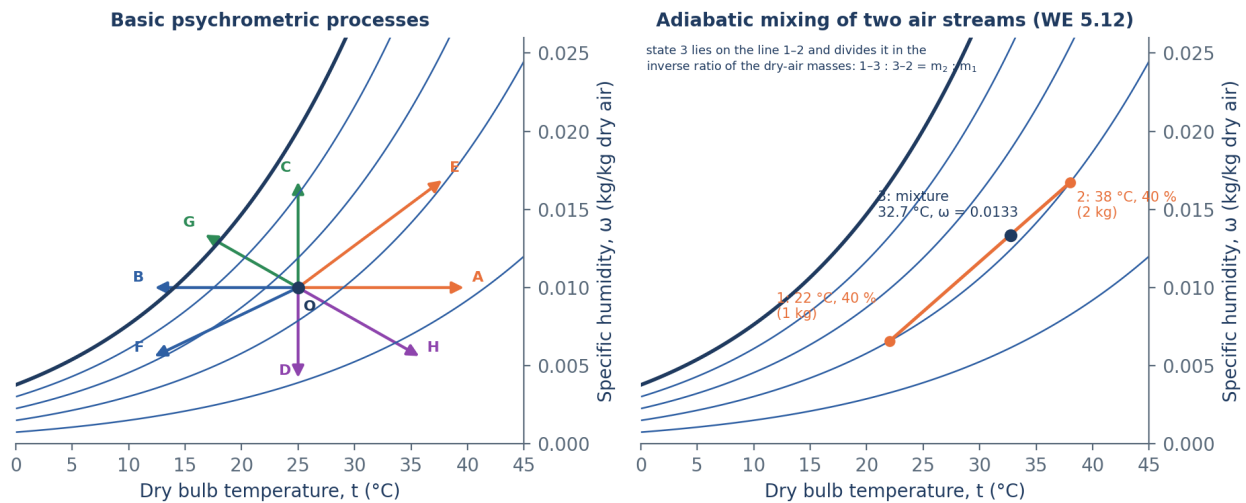


Figure 5.4 (Left) The basic psychrometric processes from a state O: A sensible heating, B sensible cooling, C humidification, D dehumidification, E heating with humidification, F cooling with dehumidification, G evaporative cooling (constant wet bulb), H chemical dehumidification. (Right) Adiabatic mixing of two streams – the mixed state lies on the straight line joining them (Worked Example 5.12).

5.1 Sensible heating and sensible cooling

Passing air over a heating coil (hot water, steam or electric) whose surface is hotter than the air, or over a cooling coil whose surface temperature is *above the dew point* of the air, changes the temperature without adding or removing moisture: ω is constant and the process is a **horizontal line** on the chart (O–A heating, O–B cooling). The heat transferred per kg of dry air is

$$q = h_2 - h_1 = (c_{pa} + 1.88\omega)(t_2 - t_1) \approx 1.02(t_2 - t_1) \quad (\text{sensible heating: } q > 0; \text{ sensible cooling: } q < 0)$$

The quantity $c_{pa} + 1.88\omega \approx 1.02 \text{ kJ/kg K}$ is the *humid specific heat*. In sensible heating the relative humidity falls (winter heating makes rooms feel dry); in sensible cooling it rises, reaching 100 % if the air is cooled to its dew point.

Sensible heat factor. In any process the total heat is split into the sensible part (change of DBT at constant ω) and the latent part (change of ω at constant DBT). The **sensible heat factor (SHF)** is

$$\text{SHF} = q_{\text{sensible}} / (q_{\text{sensible}} + q_{\text{latent}}) = (h_a - h_1) / (h_2 - h_1) \quad (\text{a horizontal process has SHF} = 1)$$

where h_a is the enthalpy at the corner state (t_2, ω_1); the SHF fixes the slope of the process line on the chart (the chart's protractor gives it directly).

By-pass factor. No coil brings all the air to its surface temperature: a fraction of the air passes through untouched. For a coil of surface temperature t_c with air entering at t_1 and leaving at t_2 , the **by-pass factor** is the fraction of air that by-passes the coil unchanged,

$$\text{BPF} = (t_2 - t_c) / (t_1 - t_c) \quad \text{contact factor} = 1 - \text{BPF} = (t_1 - t_2) / (t_1 - t_c)$$

BPF decreases with more rows of tubes, closer fin spacing and lower air velocity (typical values 0.05–0.3). The same definition, with the apparatus dew point in place of t_c , applies to cooling-and-dehumidifying coils.

5.2 Cooling and dehumidification

When moist air is cooled by a coil whose surface temperature is *below its dew point*, water vapour condenses on the coil: the air is cooled *and* dried (Figure 5.5, top). This is the process of a summer air

conditioner. The condensed water drains from the coil at approximately the exit temperature. Per kg of dry air, the mass and energy balances give

$$\text{moisture condensed } m_w = \omega_1 - \omega_2 \quad \text{heat removed } q = (h_1 - h_2) - (\omega_1 - \omega_2)h_{f2} \approx h_1 - h_2$$

the enthalpy of the condensate $(\omega_1 - \omega_2)h_{f2}$ being only a few per cent of the total and often neglected. The total heat removed splits into **sensible heat** $q_s = 1.02(t_1 - t_2)$ (from 1 to the corner state a at t_2, ω_1) and **latent heat** $q_L = (\omega_1 - \omega_2)h_{fg} \approx 2500(\omega_1 - \omega_2)$ (from a to 2); $\text{SHF} = q_s/(q_s + q_L)$. On the chart the process is a straight line from 1 towards the **apparatus dew point (ADP)** – the state of saturated air at the coil surface temperature – and the actual exit state 2 lies on this line at a distance fixed by the by-pass factor, $\text{BPF} = (t_2 - t_{\text{ADP}})/(t_1 - t_{\text{ADP}})$ (Figure 5.5, top right). With $\text{BPF} = 0$ the air leaves saturated at the ADP. For the coil to dehumidify, the ADP must be below the dew point of the entering air; the lower the ADP, the more moisture is removed but the more the air must be reheated afterwards if the room temperature is not to be too low (Worked Example 5.11).

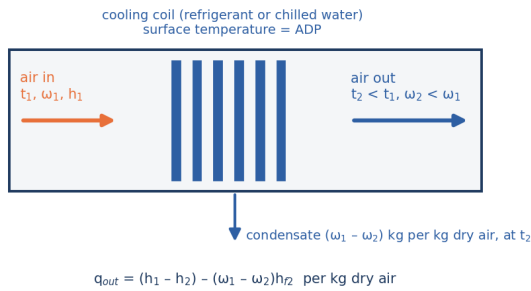
5.3 Heating and humidification

In winter, outdoor air is cold and, after heating, too dry for comfort; it is heated in a coil and then moisture is added in a humidifier (Figure 5.5, bottom). With steam injection the added vapour brings its own enthalpy, so the temperature is little changed and the humidification line 2–3 is nearly vertical; with a water spray (air washer) the latent heat comes from the air itself, which cools along a wet-bulb line (evaporative cooling, Section 5.5). The mass and energy balances for the two steps, per kg of dry air, are

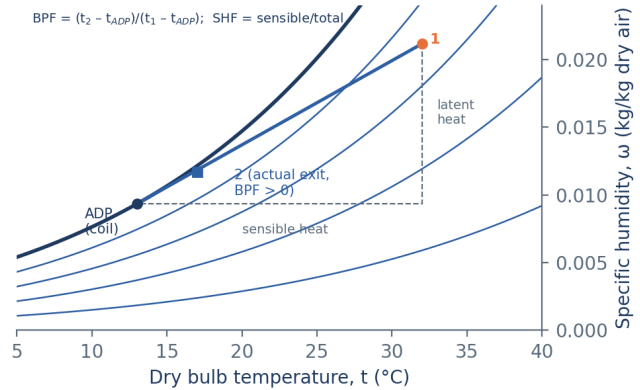
$$\text{heating 1–2: } \omega_2 = \omega_1, \quad q = h_2 - h_1 \quad \text{humidification 2–3: } m_w = \omega_3 - \omega_2, \quad h_2 + (\omega_3 - \omega_2)h_w = h_3$$

where h_w is the enthalpy of the injected steam (h_g at the steam pressure) or spray water (h_f at its temperature). If the heating and humidification are combined in a single apparatus (state 1 to state 3 directly) the total heat is $q = (h_3 - h_1) - (\omega_3 - \omega_1)h_w$, the sensible part is $1.02(t_3 - t_1)$ and the latent part $2500(\omega_3 - \omega_1)$, so the SHF of the combined process is $1.02(t_3 - t_1)/[1.02(t_3 - t_1) + 2500(\omega_3 - \omega_1)]$ and the line 1–3 slopes upward to the right (O–E in Figure 5.4). Worked Examples 5.9 and 5.10 treat steam and water-spray humidification.

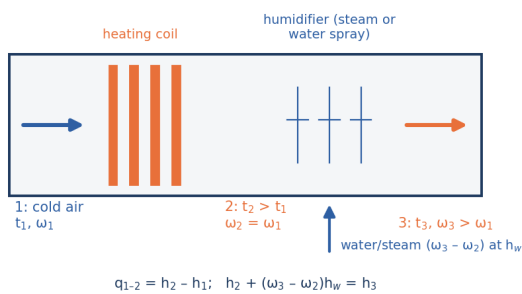
Cooling and dehumidification - cooling coil



... on the chart (with ADP and by-pass factor)



Heating and humidification - heater + humidifier



... on the chart (WE 5.9: winter air conditioner)

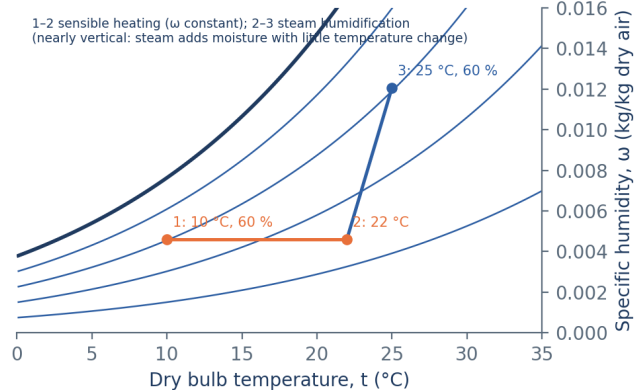


Figure 5.5 (Top) Cooling and dehumidification: the cooling coil with its condensate drain, and the process on the chart with the apparatus dew point and the by-pass factor. (Bottom) Heating and humidification: heating coil followed by a humidifier, and the winter air conditioner of Worked Example 5.9 on the chart.

5.4 Humidification and dehumidification alone

Humidification is the addition of water vapour to air (O–C, ω rises at nearly constant DBT when steam is used); **dehumidification** is its removal (O–D). Dehumidification without cooling is achieved with **chemical dehumidifiers** – silica gel or lithium chloride absorb the vapour and release its latent heat as sensible heat, so the air becomes hotter and drier along a line of nearly constant enthalpy (O–H, heating with dehumidification, the reverse of evaporative cooling). Cooling with dehumidification (Section 5.2) is the usual method in air conditioning.

5.5 Evaporative cooling

When unsaturated air passes through a spray of recirculated water or over a wetted pad (a **desert cooler** or air washer), water evaporates into the air, taking its latent heat from the air, which is cooled while its humidity rises. With no external heat transfer and the water at the wet bulb temperature, the process is adiabatic saturation: the air moves along a **constant wet bulb (constant enthalpy) line** towards the saturation curve (O–G in Figure 5.4), and the lowest temperature attainable is the wet bulb temperature. The effectiveness of an evaporative cooler is $(t_1 - t_2) / (t_1 - t_{wb})$, typically 0.8–0.9. Evaporative cooling is cheap and effective in hot, dry climates (large wet bulb depression) but useless in humid climates, where the wet bulb temperature is close to the dry bulb temperature. The same process cools the water in a **cooling tower**, where the air leaves nearly saturated and the water is cooled towards the wet bulb temperature of the entering air.

Cooling with dehumidification and evaporative cooling compared. In cooling-and-dehumidification the air loses moisture, its enthalpy falls, and heat is removed by a coil colder than the dew point; the process line slopes *downwards* to the left. In evaporative cooling the air *gains* moisture, its enthalpy is essentially unchanged, no external cooling is used, and the line runs *upwards* to the left along a wet-bulb line. The first lowers both DBT and ω (comfort cooling); the second lowers DBT but raises ω , and cannot go below the wet bulb temperature.

5.6 Adiabatic mixing of two streams of moist air

In air-conditioning plant, return air from the room is mixed with fresh outdoor air before treatment. For two streams mixing adiabatically without work (Figure 5.4, right), the balances for dry air, water and energy are

$$\dot{m}_{a1} + \dot{m}_{a2} = \dot{m}_{a3} \quad \dot{m}_{a1}\omega_1 + \dot{m}_{a2}\omega_2 = \dot{m}_{a3}\omega_3 \quad \dot{m}_{a1}h_1 + \dot{m}_{a2}h_2 = \dot{m}_{a3}h_3$$

Eliminating $\dot{m}_{a3}$ and dividing, $(\omega_3 - \omega_1)/(\omega_2 - \omega_3) = (h_3 - h_1)/(h_2 - h_3) = \dot{m}_{a2}/\dot{m}_{a1}$. Since ω and h are (very nearly) linear coordinates of the chart, this shows that **the mixed state 3 lies on the straight line joining states 1 and 2 and divides it in the inverse ratio of the dry-air masses** – the lever rule: $1-3 : 3-2 = \dot{m}_{a2} : \dot{m}_{a1}$. The dry bulb temperature of the mixture is obtained from h_3 and ω_3 ($t_3 = (h_3 - 2501\omega_3)/(1.005 + 1.88\omega_3)$) and is close to, but not exactly, the mass-weighted mean of t_1 and t_2 . If the two streams are near saturation the mixed state may fall above the saturation curve – fog – which is why the breath is visible on a cold day.

KEY POINT On the chart: sensible heating/cooling – horizontal; humidification by steam – nearly vertical; cooling and dehumidification – straight line towards the ADP on the saturation curve; evaporative cooling and water-spray humidification – along a wet-bulb (constant- h) line; adiabatic mixing – on the straight line between the two states. Every process is solved by the same three balances: dry air, water and energy.

5.7 Worked examples on psychrometric processes

WORKED EXAMPLE 5.9 — Winter air conditioner – heating followed by steam humidification

Problem. A winter air conditioner takes outside air at 10 °C, 100 kPa and 60 % relative humidity at 50 m³/min, heats it to 22 °C and then humidifies it with injected steam so that it enters the room at 25 °C and 60 % relative humidity. Show the processes on the chart and find (i) the mass flow rate of dry air, (ii) the rate of heat transfer in the heating section and (iii) the rate at which steam is added.

Given: State 1: $t_1 = 10$ °C, $\phi_1 = 0.60$, $p = 100$ kPa, $\dot{V}_1 = 50$ m³/min; $p_s(10$ °C) = 1.228 kPa. State 2: $t_2 = 22$ °C, $\omega_2 = \omega_1$. State 3: $t_3 = 25$ °C, $\phi_3 = 0.60$; $p_s(25$ °C) = 3.170 kPa.

Formula: $p_v = \phi p_s$; $\omega = 0.622 p_v / (p - p_v)$; $v_1 = R_a T_1 / (p - p_{v1})$; $\dot{m}_a = \dot{V}_1 / v_1$; $h = 1.005t + \omega(2501 + 1.88t)$; $\dot{Q} = \dot{m}_a(h_2 - h_1)$; $\dot{m}_{\text{steam}} = \dot{m}_a(\omega_3 - \omega_2)$.

Solution.

State 1: $p_{v1} = 0.60 \times 1.228 = 0.737$ kPa; $\omega_1 = 0.622 \times 0.737 / (100 - 0.737) = 0.4584/99.26 = 0.00462$ kg/kg; $v_1 = 0.287 \times 283.15/99.26 = 0.8187$ m³/kg dry air; $h_1 = 1.005 \times 10 + 0.00462 \times (2501 + 18.8) = 10.05 + 11.64 = 21.69$ kJ/kg dry air.

(i) $\dot{m}_a = 50/0.8187 = 61.07$ kg/min = 1.018 kg/s.

(ii) State 2 ($\omega_2 = 0.00462$, 22 °C): $h_2 = 1.005 \times 22 + 0.00462 \times (2501 + 41.4) = 22.11 + 11.75 = 33.86$ kJ/kg dry air.

$\dot{Q} = 1.018 \times (33.86 - 21.69) = 12.4$ kW.

(iii) State 3: $p_{v3} = 0.60 \times 3.170 = 1.902$ kPa; $\omega_3 = 0.622 \times 1.902 / (100 - 1.902) = 1.183/98.10 = 0.01206$ kg/kg.

$\dot{m}_{\text{steam}} = 61.07 \times (0.01206 - 0.00462) = 61.07 \times 0.00744 = 0.454$ kg/min = 27.3 kg/h.

ANSWER (i) $\dot{m}_a = 1.018$ kg/s; (ii) $\dot{Q}_{\text{heating}} = 12.4$ kW; (iii) steam added 0.454 kg/min (27.3 kg/h). On the chart (Figure 5.5, bottom right) 1–2 is horizontal at $\omega = 0.0046$ and 2–3 rises almost vertically from 22 °C to 25 °C – the 3 K temperature rise is supplied by the superheat of the steam. Heating alone would have left the room air at 25 °C with a relative humidity of only 23 %, hence the need for humidification.

WORKED EXAMPLE 5.10 — Heating followed by water-spray humidification

Problem. Saturated air at 4 °C is to be supplied to a room held at 22 °C and 55 % relative humidity. The air is first heated and then water at 12 °C is sprayed into it to give the required humidity. Find (i) the temperature to which the air must be heated and (ii) the mass of spray water required per m³ of air at room conditions. Total pressure 1.013 bar; fan power neglected.

Given: State 1: saturated at 4 °C, $p = 101.3$ kPa, $p_s(4\text{ °C}) = 0.8135$ kPa. State 3: 22 °C, $\phi = 0.55$, $p_s(22\text{ °C}) = 2.645$ kPa. Spray water at 12 °C: $h_w = h_f(12\text{ °C}) = 50.4$ kJ/kg.

Formula: $\omega = 0.622p_v/(p - p_v)$; $h = 1.005t + \omega(2501 + 1.88t)$; heating 1–2 at constant ω ; spray 2–3: $h_2 + (\omega_3 - \omega_2)h_w = h_3$ (adiabatic), from which h_2 and then $t_2 = (h_2 - 2501\omega_1)/(1.005 + 1.88\omega_1)$; water per m³ = $(\omega_3 - \omega_1)/v_3$.

Solution.

State 1: $\omega_1 = 0.622 \times 0.8135/(101.3 - 0.8135) = 0.5060/100.49 = 0.00504$ kg/kg; $h_1 = 1.005 \times 4 + 0.00504 \times (2501 + 7.5) = 4.02 + 12.64 = 16.66$ kJ/kg dry air.

State 3: $p_{v3} = 0.55 \times 2.645 = 1.455$ kPa; $\omega_3 = 0.622 \times 1.455/(101.3 - 1.455) = 0.9050/99.85 = 0.00906$ kg/kg; $h_3 = 1.005 \times 22 + 0.00906 \times (2501 + 41.4) = 22.11 + 23.03 = 45.14$ kJ/kg dry air; $v_3 = 0.287 \times 295.15/99.85 = 0.848$ m³/kg dry air.

Water added: $\omega_3 - \omega_1 = 0.00906 - 0.00504 = 0.00402$ kg/kg dry air.

(i) Energy balance of the spray chamber: $h_2 = h_3 - (\omega_3 - \omega_2)h_w = 45.14 - 0.00402 \times 50.4 = 45.14 - 0.20 = 44.94$ kJ/kg dry air.

$t_2 = (44.94 - 2501 \times 0.00504)/(1.005 + 1.88 \times 0.00504) = (44.94 - 12.61)/1.0145 = 32.33/1.0145 = 31.9$ °C.

(ii) Spray water per m³ of room air = $0.00402/0.848 = 0.00474$ kg/m³ = 4.7 g per m³.

ANSWER (i) The air must be heated from 4 °C to about 32 °C; (ii) 4.7 g of water is sprayed per m³ of air at room conditions. Because the spray water carries almost no enthalpy, the spray process 2–3 is nearly a constant-enthalpy (wet-bulb) line and the air cools from 32 °C to 22 °C while its humidity rises (Figure 5.6, right); the heater must therefore supply the whole enthalpy rise, $q = h_2 - h_1 = 44.94 - 16.66 = 28.3$ kJ/kg dry air, including the latent heat that is later taken up by the spray.

WORKED EXAMPLE 5.11 — Summer air conditioning – cooling and dehumidification with reheat

Problem. An air-conditioning plant is designed for outdoor air at 32 °C and 70 % relative humidity, and an indoor condition of 24 °C and 50 % relative humidity, with 3 m³/s of free (outdoor) air. The air is cooled and dehumidified in a coil whose apparatus dew point is 13 °C (leaving the coil saturated) and is then reheated to the indoor temperature. Show the processes on the chart and find (i) the cooling-coil capacity in tonnes of refrigeration, (ii) the heating-coil capacity in kW and (iii) the water removed in kg/s. Explain why reheating is needed. Take $p = 101.325$ kPa and 1 TR = 3.517 kW.

Given: Outdoor O: 32 °C, $\phi = 0.70$, $p_s(32) = 4.759$ kPa. Coil exit A: saturated at 13 °C, $p_s(13) = 1.498$ kPa. Indoor I: 24 °C, $\phi = 0.50$, $p_s(24) = 2.986$ kPa. $\dot{V}_O = 3$ m³/s.

Formula: $\omega = 0.622p_v/(p - p_v)$; $h = 1.005t + \omega(2501 + 1.88t)$; $v = R_a T/(p - p_v)$; $\dot{m}_a = \dot{V}_O/v_O$; $\dot{Q}_{cool} = \dot{m}_a(h_O - h_A)$ (condensate enthalpy neglected); $\dot{Q}_{heat} = \dot{m}_a(h_I - h_A)$; $\dot{m}_w = \dot{m}_a(\omega_O - \omega_A)$.

Solution.

Outdoor: $p_{vO} = 0.70 \times 4.759 = 3.331$ kPa; $\omega_O = 0.622 \times 3.331/(101.325 - 3.331) = 2.072/97.99 = 0.02115$ kg/kg; $h_O = 1.005 \times 32 + 0.02115 \times (2501 + 60.2) = 32.16 + 54.17 = 86.3$ kJ/kg dry air; $v_O = 0.287 \times 305.15/97.99 = 0.894$ m³/kg dry air.

$\dot{m}_a = 3/0.894 = 3.357$ kg/s.

Coil exit (ADP, saturated at 13 °C): $\omega_A = 0.622 \times 1.498/(101.325 - 1.498) = 0.9318/99.83 = 0.00933$ kg/kg; $h_A = 1.005 \times 13 + 0.00933 \times (2501 + 24.4) = 13.07 + 23.56 = 36.6$ kJ/kg dry air.

Indoor: $p_{v1} = 0.50 \times 2.986 = 1.493 \text{ kPa}$; $\omega_1 = 0.622 \times 1.493 / (101.325 - 1.493) = 0.00930 \text{ kg/kg}$ – the same as ω_A (the ADP of 13°C is the dew point of the indoor air, 12.9°C), so reheating at constant ω from A reaches the indoor state: $h_1 = 1.005 \times 24 + 0.00933 \times (2501 + 45.1) = 24.12 + 23.76 = 47.9 \text{ kJ/kg dry air}$.

(i) $\dot{Q}_{\text{cool}} = 3.357 \times (86.3 - 36.6) = 166.8 \text{ kW} = 166.8/3.517 = 47.4 \text{ TR}$.

(ii) $\dot{Q}_{\text{heat}} = 3.357 \times (47.9 - 36.6) = 37.9 \text{ kW}$.

(iii) $\dot{m}_w = 3.357 \times (0.02115 - 0.00933) = 0.0397 \text{ kg/s}$ (143 kg/h).

ANSWER (i) Cooling coil $167 \text{ kW} \approx 47 \text{ TR}$; (ii) reheater 38 kW ; (iii) 0.040 kg/s of water removed. On the chart (Figure 5.6, left) O–A is the cooling-and-dehumidifying line ending on the saturation curve at 13°C and A–I the horizontal reheat line. Reheating is necessary because the air must be cooled to 13°C to condense out enough moisture to reach $\omega = 0.0093$ (the indoor humidity), but air supplied at 13°C would overcool the room; heating it back to 24°C at constant ω gives exactly 24°C and 50% (in a real installation the room's own sensible-heat gain does part of this reheating). The 38 kW of reheat is an energy penalty inherent in controlling humidity by over-cooling.

WORKED EXAMPLE 5.12 — Adiabatic mixing of two air streams

Problem. Air at 22°C and 40% relative humidity is mixed adiabatically with air at 38°C and 40% relative humidity in the ratio 1 kg of the former to 2 kg of the latter (dry-air basis) at 101.325 kPa . Find the specific humidity, enthalpy, dry bulb temperature and relative humidity of the mixture, and show that the mixed state lies on the straight line joining the two initial states on the chart.

Given: Stream 1: 22°C , $\phi = 0.40$, $\dot{m}_{a1} = 1 \text{ kg}$, $p_s(22) = 2.645 \text{ kPa}$. Stream 2: 38°C , $\phi = 0.40$, $\dot{m}_{a2} = 2 \text{ kg}$, $p_s(38) = 6.632 \text{ kPa}$.

Formula: ω , h of each stream; $\omega_3 = (\dot{m}_{a1}\omega_1 + \dot{m}_{a2}\omega_2)/(\dot{m}_{a1} + \dot{m}_{a2})$; h_3 likewise; $t_3 = (h_3 - 2501\omega_3)/(1.005 + 1.88\omega_3)$; $p_{v3} = \omega_3 p / (0.622 + \omega_3)$; $\phi_3 = p_{v3}/p_s(t_3)$.

Solution.

Stream 1: $p_{v1} = 0.40 \times 2.645 = 1.058 \text{ kPa}$; $\omega_1 = 0.622 \times 1.058 / (101.325 - 1.058) = 0.6581/100.27 = 0.00656 \text{ kg/kg}$; $h_1 = 1.005 \times 22 + 0.00656 \times (2501 + 41.4) = 22.11 + 16.68 = 38.79 \text{ kJ/kg dry air}$.

Stream 2: $p_{v2} = 0.40 \times 6.632 = 2.653 \text{ kPa}$; $\omega_2 = 0.622 \times 2.653 / (101.325 - 2.653) = 1.650/98.67 = 0.01672 \text{ kg/kg}$; $h_2 = 1.005 \times 38 + 0.01672 \times (2501 + 71.4) = 38.19 + 43.01 = 81.20 \text{ kJ/kg dry air}$.

Mixture: $\omega_3 = (1 \times 0.00656 + 2 \times 0.01672)/3 = 0.04000/3 = 0.01333 \text{ kg/kg dry air}$.

$h_3 = (1 \times 38.79 + 2 \times 81.20)/3 = 201.19/3 = 67.06 \text{ kJ/kg dry air}$.

$t_3 = (67.06 - 2501 \times 0.01333)/(1.005 + 1.88 \times 0.01333) = (67.06 - 33.34)/1.0301 = 33.72/1.0301 = 32.7^\circ\text{C}$.

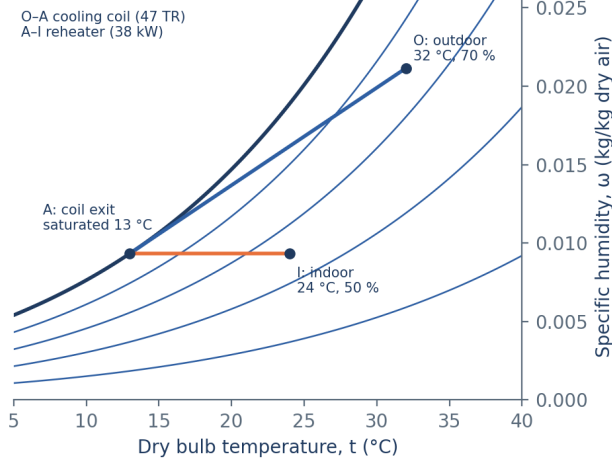
$p_{v3} = 0.01333 \times 101.325 / (0.622 + 0.01333) = 1.351/0.6353 = 2.126 \text{ kPa}$; $p_s(32.7^\circ\text{C}) = 4.95 \text{ kPa}$; $\phi_3 = 2.126/4.95 = 0.43 = 43\%$.

Straight-line check: $(\omega_3 - \omega_1)/(\omega_2 - \omega_3) = (0.01333 - 0.00656)/(0.01672 - 0.01333) =$

$0.00677/0.00339 = 2.0 = \dot{m}_{a2}/\dot{m}_{a1} \checkmark$, and $(h_3 - h_1)/(h_2 - h_3) = 28.27/14.14 = 2.0 \checkmark$: state 3 divides the line 1–2 in the ratio $2 : 1$, i.e. it lies on the line, twice as far from 1 as from 2 (Figure 5.4, right).

ANSWER $\omega_3 = 0.0133 \text{ kg/kg dry air}$; $h_3 = 67.1 \text{ kJ/kg dry air}$; $t_3 = 32.7^\circ\text{C}$; $\phi_3 = 43\%$. The mixture temperature (32.7°C) is close to the mass-weighted mean (32.67°C), and the relative humidity of the mixture (43%) is slightly *higher* than that of either stream (40%) because the saturation curve is convex – the reason fog can form when two nearly saturated streams mix.

Summer AC: cooling-dehumidification + reheat (WE 5.11)



Heating then water-spray humidification (WE 5.10)

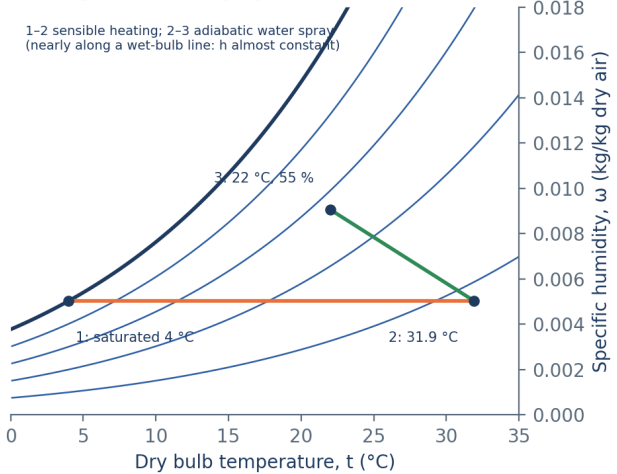


Figure 5.6 (Left) Summer air conditioning of Worked Example 5.11: cooling and dehumidification to the apparatus dew point followed by reheating. (Right) Heating followed by water-spray humidification (Worked Example 5.10).

6. Simple applications

- **Summer air conditioning.** Outdoor air (often mixed with return air) is cooled and dehumidified by a coil whose ADP is below the room dew point, reheated if necessary and supplied to the room, where it absorbs the sensible and latent heat gains (people, lights, solar gain, infiltration) along the room's sensible-heat-factor line back to the room state (Worked Example 5.11). The cooling-coil load, in kW or tonnes of refrigeration, fixes the size of the refrigeration plant.
- **Winter air conditioning.** Outdoor air is heated, humidified by steam or water spray, and supplied to the room (Worked Examples 5.9 and 5.10); the humidifier restores the relative humidity that heating alone would reduce to an uncomfortable 20–30 %.
- **Evaporative (desert) coolers.** In hot, dry climates air is cooled by 10–15 K by passing it through wetted pads, with no refrigeration plant (Section 5.5).
- **Cooling towers.** Warm condenser water is sprayed through an upward stream of air; evaporation of about 1–2 % of the water cools the rest towards the wet bulb temperature of the air, which leaves nearly saturated – the largest-scale evaporative cooling process in engineering.
- **Drying.** Heated air of low relative humidity is passed over the material (grain, timber, textiles, paper); the air picks up moisture along a wet-bulb line and its exit humidity, read from the chart, gives the drying capacity per kg of air.
- **Comfort.** Human comfort depends on DBT, relative humidity and air movement; the "comfort zone" on the chart lies roughly between 20 and 26 °C and 30 to 60 % relative humidity, and the whole purpose of the processes above is to bring the supply air to a state that will hold the room within it.

7. Practice problems (answers given)

No.	Problem	Answer
P1	Air is 79 % N ₂ and 21 % O ₂ by volume (M = 28 and 32). Find its molar mass, gas constant and the mass fractions.	M = 28.84 kg/kmol; R = 0.288 kJ/kg K; x _{N₂} = 0.767, x _{O₂} = 0.233
P2	A mixture of 2 kmol CO ₂ and 3 kmol N ₂ is at 300 kPa. Find the partial pressures and the mass fractions.	p _{CO₂} = 120 kPa, p _{N₂} = 180 kPa; x _{CO₂} = 0.512, x _{N₂} = 0.488
P3	One kmol of O ₂ and one kmol of N ₂ , each at 100 kPa and 25 °C, mix adiabatically to 100 kPa. Find the entropy generated.	ΔS = -R _u Σn _i ln y _i = 2 × 8.3145 × ln 2 = 11.53 kJ/K
P4	Find ω, h, v and the dew point of air at 25 °C, 50 % RH and 101.325 kPa (p _s = 3.170 kPa).	ω = 0.00988 kg/kg; h = 50.3 kJ/kg; v = 0.858 m ³ /kg; t _{dp} = 13.9 °C

No.	Problem	Answer
P5	Air at 15 °C and 60 % RH is heated sensibly to 30 °C. Find the heat added per kg dry air and the final relative humidity.	$q = 15.3 \text{ kJ/kg dry air}$; $\phi_2 = 24 \%$
P6	Air at 30 °C, 60 % RH is cooled to saturation at 14 °C. Find the moisture removed and the heat removed per kg dry air (neglect condensate enthalpy).	0.0061 kg/kg ; $31.9 \text{ kJ/kg dry air}$
P7	Air at 38 °C and 20 % RH passes through an evaporative cooler of effectiveness 0.9. Find the exit temperature ($t_{wb} = 20.8 \text{ °C}$).	22.5 °C
P8	Equal dry-air masses at (15 °C, 80 % RH) and (30 °C, 50 % RH) mix adiabatically at 101.325 kPa. Find ω , h , t and ϕ of the mixture.	$\omega = 0.0109$; $h = 50.4 \text{ kJ/kg}$; $t = 22.5 \text{ °C}$; $\phi = 64 \%$

8. Summary and formula sheet

- Composition:** $x_i = m_i/m$ (mass fraction), $y_i = n_i/n$ (mole fraction); $x_i = y_i M_i/M$; $M = \sum y_i M_i = 1/\sum (x_i/M_i)$; $R = R_u/M = \sum x_i R_i$. **Dalton:** $p = \sum p_i$, $p_i = y_i p$ (components at V , T). **Amagat:** $V = \sum V_i$, $V_i = y_i V$ (components at p , T). For ideal gases $p_i/p = V_i/V = y_i$.
- Mixture properties:** $\rho = \sum \rho_i = p/(RT)$; $c_p = \sum x_i c_{p,i}$, $c_v = \sum x_i c_{v,i}$; $\Delta U = \sum m_i c_{v,i} \Delta T$; $\Delta H = \sum m_i c_{p,i} \Delta T$; $\Delta S = \sum m_i [c_{p,i} \ln(T_2/T_1) - R_i \ln(p_{i2}/p_{i1})]$ (partial pressures); **entropy of mixing** $\Delta S_{\text{mix}} = -R_u \sum n_i \ln y_i = \sum m_i R_i \ln(V/V_i) > 0$; $G = \sum n_i [\bar{g}_i(T, p) + R_u T \ln y_i]$.
- Moist air:** $p = p_a + p_v$; $\omega = 0.622 p_v/(p - p_v)$; $\phi = p_v/p_s$; $\mu = \omega/\omega_s = \phi(p - p_s)/(p - p_v)$; $h = 1.005t + \omega(2501 + 1.88t) \text{ kJ/kg dry air}$; $v = 0.287T/(p - p_v)$; $p_a = 1/v$, $p_v = p_v/(0.4615T)$; $t_{dp} = T_{\text{sat}}(p_v)$. DBT, WBT, DPT coincide for saturated air.
- Adiabatic saturation / wet bulb:** $\omega_1 = [c_{pa}(t_2 - t_1) + \omega_2 h_{fg2}]/(h_{g1} - h_{f2})$; nearly constant h ; $t_{wb} \approx$ adiabatic saturation temperature for air–water vapour.
- Chart:** DBT (x) vs ω (y): saturation curve, ϕ curves, WBT lines, h lines, v lines; two properties fix the state.
- Processes (per kg dry air):** sensible heating/cooling $q = h_2 - h_1 \approx 1.02\Delta t$, ω constant; SHF = $q_s/(q_s + q_L)$; BPF = $(t_2 - t_c)/(t_1 - t_c)$. Cooling–dehumidification: $m_w = \omega_1 - \omega_2$, $q = (h_1 - h_2) - (\omega_1 - \omega_2)h_{f2}$, line to the ADP. Heating–humidification: $q = h_2 - h_1$; $h_2 + (\omega_3 - \omega_2)h_w = h_3$. Evaporative cooling: constant wet bulb, limit t_{wb} , effectiveness $(t_1 - t_2)/(t_1 - t_{wb})$. Mixing: $\omega_3 = (\dot{m}_1 \omega_1 + \dot{m}_2 \omega_2)/(\dot{m}_1 + \dot{m}_2)$, h_3 likewise; state 3 on the line 1–2, lever rule $\dot{m}_2/\dot{m}_1 = (\omega_3 - \omega_1)/(\omega_2 - \omega_3)$.

9. Review questions

9.1 Short-answer (two-mark) questions

- Define the mole fraction and the mass fraction of a component of a gas mixture.
- State Dalton's law of partial pressures.
- State Amagat's law of partial volumes.
- What is the by-pass factor of a cooling coil?
- Write the expression for the gas constant of an ideal-gas mixture.
- Write the expression for the density of an ideal-gas mixture.
- Express the specific heats of a gas mixture in terms of the mass fractions of its components.
- Write the expression for the change in internal energy of an ideal-gas mixture.
- Write the expression for the change in enthalpy of an ideal-gas mixture.
- Write the expression for the entropy generated when two ideal gases mix adiabatically.
- What are humidification and dehumidification?
- How do the sensible heat and the latent heat of moist air differ?
- What is psychrometry? Mention two applications.
- Define the dry bulb, wet bulb and dew point temperatures.

15. Define specific humidity and express it in terms of partial pressures.
16. Define relative humidity and write its expression.
17. How does specific humidity differ from relative humidity?
18. Define the degree of saturation.
19. Define the enthalpy of moist air and write its expression per kg of dry air.
20. What is the adiabatic saturation process? How is the adiabatic saturation temperature related to the wet bulb temperature?
21. Name the families of lines on a psychrometric chart.
22. Define the sensible heat factor and write its expression.
23. Sketch sensible heating and sensible cooling on the psychrometric chart.
24. How does cooling with dehumidification differ from evaporative cooling?
25. Explain adiabatic mixing of two air streams on the psychrometric chart.
26. Why does the relative humidity fall when air is heated at constant specific humidity?
27. What is the apparatus dew point of a cooling coil?
28. Why can an evaporative cooler not cool air below its wet bulb temperature?
29. Why is reheating used after cooling and dehumidification?
30. State the Gibbs–Dalton law.

9.2 Long-answer (thirteen-mark) questions

1. With a sketch of the apparatus and the process on the psychrometric chart, explain cooling and dehumidification of moist air; derive the expressions for the sensible, latent and total heat removed and the moisture condensed per kg of dry air, and define the sensible heat factor and the by-pass factor of the coil.
2. With a sketch of the apparatus and the process on the chart, explain heating and humidification of moist air; from the mass and energy balances derive the heat added, the vapour added per kg of dry air and the sensible heat factor.
3. A gas mixture is 55 % N_2 , 30 % O_2 and 15 % CO_2 by mass. Find the mole fractions, the molar mass and gas constant of the mixture, the partial pressures at 250 kPa and 320 K, and the volume of 4 kg of the mixture.
4. A rigid insulated tank has 3 kg of N_2 at 250 kPa, 30 °C in one compartment and 2 kg of CO_2 at 120 kPa, 30 °C in the other; the partition is removed. Find the tank volume, the final pressure, the partial pressures and the entropy generated ($R_{N_2} = 0.2968$, $R_{CO_2} = 0.1889$ kJ/kg K).
5. Atmospheric air at 1.01325 bar has DBT 32 °C and WBT 24 °C. Using the chart find the specific humidity, relative humidity, dew point, enthalpy and specific volume.
6. Moist air at 1.01325 bar, 36 °C and 45 % RH: find from the chart the specific humidity, dew point, wet bulb temperature, enthalpy and specific volume; if it is cooled to 12 °C, determine whether condensation occurs and the moisture removed per kg of dry air.
7. Moist air at 1.01325 bar has DBT 30 °C and DPT 18 °C. Find from the chart the specific humidity, relative humidity, wet bulb temperature, enthalpy and specific volume, and verify ω and ϕ by calculation with p_s at 18 °C and 30 °C.
8. Air at 1.0132 bar has DBT 34 °C and WBT 25 °C. Calculate the vapour pressure, specific humidity, dew point, relative humidity, degree of saturation, densities of dry air and vapour, and the enthalpy per kg of dry air.
9. A winter air conditioner takes outside air at 10 °C, 100 kPa, 60 % RH at 50 m³/min, heats it to 22 °C and humidifies it with steam to 25 °C and 60 % RH. Show the processes on the chart and find the dry-air flow, the heating rate and the steam added.
10. Air enters an adiabatic saturator at 32 °C and leaves saturated at 22 °C at 101.325 kPa. Find the humidity ratio, relative humidity and dew point of the entering air and show the process on the chart.
11. Saturated air at 4 °C is heated and then sprayed with water at 12 °C to give room air at 22 °C and 55 % RH at 1.013 bar. Find the temperature to which the air is heated and the spray water per m³ of room air.

12. An air-conditioning plant handles $3 \text{ m}^3/\text{s}$ of outdoor air at 32°C , 70 % RH; the indoor condition is 24°C , 50 % RH; the coil ADP is 13°C and the air is reheated after the coil. Show the processes on the chart and find the cooling-coil capacity in TR, the heating-coil capacity in kW and the water removed in kg/s; explain why reheating is needed.
13. Air at 22°C , 40 % RH mixes adiabatically with twice its dry-air mass of air at 38°C , 40 % RH. Find ω , h , DBT and ϕ of the mixture and show that the mixed state lies on the line joining the initial states.
14. A room $5 \text{ m} \times 4 \text{ m} \times 3 \text{ m}$ contains air at 28°C , 100 kPa and 60 % RH. Find the partial pressure of dry air, the specific humidity, the enthalpy per kg of dry air and the masses of dry air and vapour.
15. Describe the psychrometric chart – its coordinates and the families of lines on it – and explain how each of the following is represented: sensible heating, sensible cooling, cooling with dehumidification, heating with humidification, evaporative cooling and adiabatic mixing.
16. Derive the expressions for the entropy change of an ideal-gas mixture and for the entropy of mixing of ideal gases initially at the same pressure and temperature, and show that the entropy of mixing depends only on the mole numbers.

Text: P. K. Nag, "Engineering Thermodynamics", 6th Edn., McGraw-Hill — Chapters 10 and 16. References: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.; R. K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edn.; C. P. Arora, "Refrigeration and Air Conditioning". Water properties: IAPWS-97.